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Descent in Probability Theory: the first steps downward

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Coarse-graining, forming quotients by dropping distinctions, is a unifying idea across mathematics: identifying the endpoints of an interval yields a circle; groups are conveniently presented as quotients of free ones; sheaves and stacks emerge from gluing local data. This idea is also central to probability, where "observing" a random variable similarly quotients a sample space via the sigma-algebra it generates (and is crucial for modeling randomness as "ignorance"). Yet this quotienting procedure, in probability, has so far lacked a systematic categorical treatment.

We develop a descent theory for probability that makes this intuition precise, while respecting probabilistic practice as much as possible. On the category theory side, the theory parallels classical descent, but diverges in a few ways due to the presence of stochastic dependence (correlations). On the probability side, it unifies the three core concepts of measurability, disintegration and stochastic dominance, within a single framework, providing conceptual understanding of the relationships between random variables, statistical experiments, and inference procedures.

Partly based on joint work with Dario Stein and with Mika Bohinen.

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References

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